Mean-anchored self-supervised reconstruction from few speckle measurements in light-field microscopy

1. Introduction

Light-field microscopy (LFM) enables volumetric fluorescence imaging by recording spatial and angular information in a single camera exposure [1,2]. A microlens array encodes light from different depths onto the detector, allowing a three-dimensional (3D) volume to be reconstructed without sequentially scanning the focal plane. This parallel acquisition is attractive for observing structures and processes distributed throughout a specimen. However, distributing detector samples between spatial and angular measurements limits the spatial detail available in conventional LFM [1]. Wave-optics-based deconvolution and virtual-scanning reconstruction have improved the recovery of this information [2,3]. The ability to resolve neighboring structures nevertheless depends on both the optical encoding and the reconstruction, motivating approaches that introduce additional spatial information into the measurements.

Speckle illumination provides such information by modulating fluorescence excitation with spatially varying intensity patterns. Blind structured illumination microscopy (blind-SIM) demonstrated that resolution can be enhanced using unknown speckle patterns under appropriate illumination conditions [4]. Theoretical work subsequently established how measurement covariance can extend the recoverable spatial-frequency support [5], while random illumination microscopy demonstrated the use of speckle statistics for super-resolved live-cell imaging [6]. In LFM, Taylor et al. used illumination-induced fluorescence fluctuations and variance-based reconstruction to improve spatial resolution and suppress background in volumetric neuronal imaging [7]. These developments establish a direct basis for combining random illumination with light-field detection. They also identify the relationship between illumination statistics and reconstructed spatial structure as a central part of the imaging problem.

The resolution benefit of speckle illumination must be considered together with its acquisition cost. A statistical reconstruction draws information from a sequence of illumination realizations, whose number affects both the available measurements and the estimation of their statistics. With only a short sequence, the measured mean and variance remain sensitive to the particular speckle patterns sampled. Under spatially uniform mean illumination, the measured mean describes the average fluorescence response, whereas fluctuations provide complementary spatial information [5,7]. Increasing the number of measurements can improve statistical estimation, but at fixed exposure and switching times it also lengthens the acquisition interval. The challenge is therefore to recover reliable fine structure from a limited sequence, so that reducing the frame count preserves the spatial information needed to distinguish neighboring features and trace continuous structures.

Learning-based reconstruction offers a way to incorporate structural priors into this limited-measurement problem. Supervised methods such as VCD-Net and HyLFM-Net have improved reconstruction throughput and image quality in LFM [8,9], although their training requires suitable reference data. Physics-driven self-supervision provides an alternative when matched high-resolution targets are difficult to obtain. SeReNet demonstrated self-supervised volumetric reconstruction based on the light-field imaging process [10], and unrolled blind-SIM combined learned reconstruction with a physical model of unknown illumination [11]. These advances motivate a reconstruction strategy that uses learned priors together with the statistics of a short speckle sequence. For speckle-illuminated LFM, the specific question addressed here is how mean, fluctuation, and individual-frame information can be combined within the light-field forward model to recover fine spatial structure at a small measurement budget.

Here, we present a mean-anchored self-supervised method for volumetric reconstruction from ten speckle-illuminated light-field measurements. A preliminary reconstruction from the measurement mean defines the output anchor, while a variance-derived reconstruction and features from the individual frames guide learned corrections. The method separates overall intensity from normalized structure and bounds the contribution of mean-consistency gradients to structural updates, providing explicit control over the combination of mean and variance constraints. Training uses disjoint input and constraint subsets without 3D ground-truth targets; inference uses only the ten input measurements. We evaluate the method using simulated targets with known lateral and axial structure, quantifying feature separation, volumetric fidelity, and variation across measurement subsets.

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